

Manzil JEE 2026

Chemistry (Practice Sheet)

Classification of Elements and Periodicity in Properties

Time duration - 60 min.

- Q1** The nuclei of elements X, Y and Z have same number of protons, but different numbers of neutrons. According to Mendeleev periodic table, the elements X, Y and Z
- belong to the same group and the same period
 - belong to different groups and different periods
 - belong to the same group and different periods
 - are isotopes, which do not have different positions
- Q2** Which of the following species have the same number of electrons in its outermost as well as penultimate shell?
- Mg^{2+}
 - O^{3-}
 - F^-
 - Ca^{2+}
- Q3** Which of the following pairs has the same size?
- Fe^{2+} , Ni^{2+}
 - Zr^{4+} , Ti^{4+}
 - Zr^{4+} , Hf^{4+}
 - Zn^{4+} , Hf^{4+}
- Q4** An element has 56 nucleons in nucleus and if it is isotonic with ${}_{30}\text{Y}^{60}$. Which group and period does it belong to?
- 8th group, 4th period
 - 14th group, 3rd period
 - 12th group, 3rd period
 - 12th group, 4th period
- Q5** In which of the following arrangements the order is NOT according to the property indicated
- against it?
- $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+ < \text{F}^-$ – increasing ionic size
 - $\text{B} < \text{C} < \text{N} < \text{O}$ - increasing first ionisation enthalpy
 - $\text{I} < \text{Br} < \text{F} < \text{Cl}$ - increasing electron gain enthalpy (with negative sign)
 - $\text{Li} < \text{Na} < \text{K} < \text{Rb}$ - increasing metallic radius
- Q6** The electron gain enthalpies of halogens in kJmol^{-1} are given below.
 $\text{F} = -332$, $\text{Cl} = -349$, $\text{Br} = -325$, $\text{I} = -295$
 The lesser negative value for F as compared to that of Cl is due to
- Strong electron-electron repulsions in the compact 2p-subshell of F.
 - Weak electron-electron repulsions in the bigger 3p-subshell of Cl.
 - Smaller electronegativity value of F than Cl.
 - 1&2 both
- Q7** The correct order of radii is
- $\text{N} < \text{Be} < \text{B}$
 - $\text{F}^- < \text{O}^{2-} < \text{N}^{3-}$
 - $\text{Na} < \text{Li} < \text{K}$
 - $\text{Fe}^{3+} < \text{Fe}^{2+} < \text{Fe}^{4+}$
- Q8** Correct order of acidic strength is
- $\text{N}_2\text{O}_5 < \text{P}_4\text{O}_{10}$
 - $\text{CO}_2 > \text{SiO}_2$
 - $\text{B}_2\text{O}_3 < \text{BeO}$
 - $\text{HClO}_3 > \text{HClO}_4$
- Q9** Series-I: $[\text{N}^{3-}, \text{O}^-, \text{Na}^+]$; Series-II: $[\text{N}^+, \text{C}^+, \text{O}^+]$ Choose the species of lowest I.P.



from Series-I and highest I.P. from Series-II respectively

- (A) N^{3-} , O^+
- (B) Na^+ , C^+
- (C) N^{3-} , N^+
- (D) O^- , C^+

- Q10** The decreasing order of second ionization potential of K, Ca, Ba is
- (A) $\text{K} > \text{Ca} > \text{Ba}$
 - (B) $\text{Ca} > \text{Ba} > \text{K}$
 - (C) $\text{Ba} > \text{K} < \text{Ca}$
 - (D) $\text{K} > \text{Ba} > \text{Ca}$
- Q11** For the formation of covalent bond, the difference in the value of electronegativities should be:
- (A) Less than 1.7
 - (B) More than 1.7
 - (C) 1.7 or more
 - (D) None of the above
- Q12** Which of the following statement is **incorrect**?
- (A) Be has higher ionisation enthalpy than B because of its stable electronic configuration.
 - (B) Electron gain enthalpy is always negative for a neutral atom.
 - (C) Depending on the element, the process of adding an electron to an atom of the element can either be endothermic or exothermic.
 - (D) Ionic radius of O^{2-} is higher than that of F^- .
- Q13** An atom with high EA generally has
- (A) Tendency to form +ve ions
 - (B) High ionization energy
 - (C) Large atomic size
 - (D) Low electron affinity
- Q14** Assign TRUE (T) or FALSE(F) for the following statements and select CORRECT option for your answer.
- I. I.P of $\text{O}_{(g)}$ is less than I.P. of $\text{O}^-(g)$
 - II. I.P of $\text{Ne}_{(g)}$ is greater than I.P. of $\text{Ne}^+(g)$
 - III. E.A. of $\text{O}^+(g)$ is greater than E.A. of $\text{O}_{(g)}$
 - IV. I.P of $\text{N}_{(g)}$ is greater than I.P. of $\text{N}^+(g)$

- (A) F, F, T, F
- (B) T, T, T, T
- (C) T, T, T, F
- (D) F, T, F, T

- Q15** The difference between electron gain enthalpies will be maximum between:
- (A) Ne and F
 - (B) Ne and Cl
 - (C) Ar and Cl
 - (D) Ar and F
- Q16** The screening effect of inner electrons of the nucleus causes
- (A) Decrease in the ionisation energy
 - (B) Increase in the ionisation energy
 - (C) No effect on the ionisation energy
 - (D) Increases in the attraction of the nucleus to the electrons
- Q17** Consider the following elements:
Cl, F, S, P
The **correct** statement regarding the above elements is:
- (A) Sulphur has the most negative electron gain enthalpy among the given elements.
 - (B) Adding an electron to 3p-orbital leads to greater repulsion than adding an electron to 2p-orbital.
 - (C) Phosphorus has the least negative electron gain enthalpy among the given elements.
 - (D) As we move from left to right across a period, the electron gain enthalpy generally becomes less negative.
- Q18** In the compound $\text{M} - \text{O} - \text{H}$, the $\text{M} - \text{O}$ bond will be broken if:
- (A) $\Delta(\text{E.N.})$ of M and O $<$ $\Delta(\text{E.N.})$ of O and H
 - (B) $\Delta(\text{E.N.})$ of M and O $=$ $\Delta(\text{E.N.})$ of O and H
 - (C) $\Delta(\text{E.N.})$ of M and O $>$ $\Delta(\text{E.N.})$ of O and H
 - (D) Cannot be predicted according $\Delta(\text{E.N.})$ data
- Q19** The ionization energy of boron is less than that of beryllium because
- (A) beryllium has a higher nuclear charge than boron



- (B) beryllium has a lower nuclear charge than boron
 (C) the outermost electron in boron occupies a 2p orbital
 (D) the 2s and 2p-orbitals of boron are degenerate

Q20 The first four I.E. values of an element are 284, 412, 656 and 3210 kJ mol⁻¹. The number of valence electrons in the element are:

- (A) one (B) two
 (C) three (D) four

Q21 Amphoteric oxide combinations are in

- (A) ZnO, K₂O, SO₃
 (B) ZnO, P₂O₅, Cl₂O₇
 (C) SnO₂, Al₂O₃, ZnO
 (D) PbO₂, SnO₂, SO₃

Q22 Which from following series of elements is correctly arranged according to their decreasing order of ionisation enthalpy (IE₁) ?

- (A) Zn > Fe > Cr > Sc
 (B) Cr > Fe > Zn > Sc
 (C) Sc > Fe > Cr > Zn
 (D) Cr > Zn > Sc > Fe

Q23 Which of the following order of ionic radii is correctly represented?

- (A) H⁺ > H⁻ > H
 (B) Na⁺ > F⁻ > O²⁻
 (C) O²⁻ > F⁻ > Na⁺
 (D) Al³⁺ > Mg²⁺ > N³⁻

Q24 Z_{eff} of Na⁺ and Na are respectively:

- (A) 1.95 and 1.22 (B) 0.85 and 0.35
 (C) 1 and 0.35 (D) 6.85 and 2.2

Q25 The first and second ionisation enthalpies (in kJ mol⁻¹) of a few elements are given below

Element	IE ₁	IE ₂
I.	2372	5251
II.	520	7300
III.	900	1760
IV.	1680	3380

Which of the above elements is likely to be a metal that forms a stable binary halide of the formula AX₂?

- (A) I (B) II
 (C) III (D) IV

Q26 The process that requires absorption of energy is:

- (A) N → N⁻
 (B) F → F⁻
 (C) Cl → Cl⁻
 (D) H → H⁻

Q27 The electronegativities of C, N, Si and P are in the order of:

- (A) P < Si < C < N
 (B) Si < P < C < N
 (C) P < Si < N < C
 (D) Si < P < N < C

Q28 Successive ionisation energies of an element A are

100 eV, 150 eV, 181 eV, 2000 eV, 2200 eV
 correct statement regarding A is:

- (A) Element A may be metal
 (B) Formula of oxide of A may be A₂O₃
 (C) Oxide of element A may be amphoteric
 (D) All are correct

Q29 Identify the group (in Modern Periodic Table) and valency of a hypothetical element having atomic number 119. If group number is x and valency is y. Give the value of x + y.

Q30 ZnO, Al₂O₃, V₂O₅, V₂O₃, Cr₂O₃, Na₂O, CaO, CrO₃, Mn₂O₇, N₂O₃, SO₂, CO₂

If the number of acidic oxides is x, number of basic oxides is y and the number of amphoteric oxides is z. Then the value of (x² + 2y + z) is _____.

Q31 If there were 9 periods in the periodic table & each orbital can have maximum 5 electrons, then how many maximum number of elements will be present in period 9 ?

Q32



An element 'X' has its electronic configuration of 'K' shell as $(n - 5)s^2$ and the total number of electrons in its outermost, penultimate and antipenultimate shell are 2, 8 and 25 respectively. Find out total number of unpaired electrons in element 'X' in its ground state.

Q33 Find out total number of representative elements in the given elements:

Zn, Mg, Na, Fe, N, F, S, B, Al

Q34 An element with IUPAC symbol 'Unq' belongs to period number 'x' and group number 'y'. The sum $x + y$ is _____.

Q35 What is the % ionic character in HI ? Given electronegativity of $H = 2.1$ and $I = 2.5$. (Give your answer to the nearest integer)

Q36 B_2O_3 , Al_2O_3 , Ga_2O_3 , In_2O_3 , Tl_2O_3 , $B(OH)_3$, $Al(OH)_3$, $Ga(OH)_3$, $In(OH)_3$, $Tl(OH)_3$.
If the number of amphoteric compound is X and basic compounds is Y, then the value of $X + Y$ is:

Q37 From the given element (s), how many have ionization energy less than chlorine atom?

Na, Mg, P, S, Ar, Al

Q38 Consider two electronic configuration of K (Potassium) given below :
(I) $K = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ (follows $n + \ell$ rule)

(II) $K = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^1$ (violates $n + \ell$ rule)

The effective nuclear charge experienced by last electron in (I) and (II) is x and y. Then find the value of $x+y$. (Nearest integer)

Q39 The electron gain enthalpy of a hypothetical element 'A' is $-3eV$ per atom. How much energy (in $kcalmol^{-1}$) is released when 10 g of 'A' are completely converted to A^- ions in gaseous state?

(Take : $1eV$ per atom = $23kcalmol^{-1}$,
Molar mass of A = 30 g)

Q40 Find out number of correct statements.

A. $N_2 > N$ (1st ionization energy)

B. $O_2 > O$ (1st ionization energy)

C. $H_2 > H$ (1st ionization energy)

D. $N_2^+ > N_2^-$ (Stability)

E. $O_2 > O$ (1st Electron affinity)

F. $H_2 > H$ (1st Electron affinity)

G. $Na_2CO_3 > Li_2CO_3$ (Thermal stability)

H. No. of co-ordinate bond in P_4O_{10} is 4

I. No. of π bond in P_4O_8 is 2



Answer Key

Q1 (D)
Q2 (D)
Q3 (C)
Q4 (A)
Q5 (B)
Q6 (D)
Q7 (B)
Q8 (B)
Q9 (A)
Q10 (A)
Q11 (A)
Q12 (B)
Q13 (B)
Q14 (A)
Q15 (B)
Q16 (A)
Q17 (C)
Q18 (C)
Q19 (C)
Q20 (C)

Q21 (C)
Q22 (A)
Q23 (C)
Q24 (D)
Q25 (C)
Q26 (A)
Q27 (B)
Q28 (D)
Q29 2
Q30 35
Q31 180
Q32 7
Q33 7
Q34 11
Q35 7
Q36 8
Q37 5
Q38 3
Q39 23
Q40 6



Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

- Elements **X, Y, and Z** have the **same number of protons** but **different numbers of neutrons**. This means they are **isotopes** of the same element.
- **Mendeleev's periodic table** was based on **atomic mass**, but he did **not assign different positions** to isotopes, even though they have different atomic masses.
- All isotopes of an element were placed **at the same position** in his table.

Video Solution:



Q2 Text Solution:

- **A. Mg^{2+}** : Neutral Mg is $[\text{Ne}] 3s^2$. Mg^{2+} loses 2 electrons $\rightarrow [\text{Ne}]$ (10 electrons). Outermost shell ($n=2$): 8 electrons. Penultimate shell ($n=1$): 2 electrons. **Not equal.**
- **B. O^{3-}** : Neutral O is $[\text{He}] 2s^2 2p^4$. O^{3-} gains 3 electrons $\rightarrow [\text{He}] 2s^2 2p^6$ (10 electrons, same as Ne). Outermost shell ($n=2$): 8 electrons. Penultimate shell ($n=1$): 2 electrons. **Not equal.**
- **C. F^-** : Neutral F is $[\text{He}] 2s^2 2p^5$. F^- gains 1 electron $\rightarrow [\text{He}] 2s^2 2p^6$ (10 electrons, same as Ne). Outermost shell ($n=2$): 8 electrons. Penultimate shell ($n=1$): 2 electrons. **Not equal.**
- **D. Ca^{2+}** : Neutral Ca is $[\text{Ar}] 4s^2$. Ca^{2+} loses 2 electrons $\rightarrow [\text{Ar}]$ (18 electrons). Outermost shell ($n=3$): 8 electrons ($3s^2 3p^6$). Penultimate shell ($n=2$): 8 electrons ($2s^2 2p^6$). **Equal.**

Video Solution:



Q3 Text Solution:

Zr^{4+} and Hf^{4+} has same size due to Lanthnoid contraction.

Video Solution:



Q4 Text Solution:

Given:

The element has **56 nucleons** $\rightarrow$ Mass number $A = 56$

It is isotonic with ${}^{60}_{30}\text{Y} \rightarrow$ Both have the **same number of neutrons**

Neutrons = $A - Z = 60 - 30 = 30$.

For the element: $A = 56$, neutrons = 30 $\Rightarrow$

$Z = A - \text{neutrons} = 56 - 30 = 26$.

$Z = 26 \rightarrow$ Iron (Fe).

- Electron configuration: $[\text{Ar}] 3d^6 4s^2$
- Valence electrons in 4s $\rightarrow$ Group 8 (old IUPAC)
- Highest energy level $n=4 \rightarrow$ 4th period

Video Solution:



Q5 Text Solution:



As we move from left to right across a period, ionisation enthalpy increases with increasing atomic number. So the order of increasing ionisation enthalpy be $B < C < N < O$. But N ($1s^2 2s^2 2p^3$) has a stable half filled electronic configuration. So, ionization enthalpy of nitrogen is greater than oxygen. So, the correct order of increasing first ionization enthalpy is $B < C < O < N$.

Video Solution:



Q6 Text Solution:

The extra electron in fluorine must squeeze into a very small, tightly packed 2p subshell, where inter-electronic repulsion is intense (statement 1). In chlorine the incoming electron enters the more diffuse 3p subshell, so repulsion is weaker (statement 2).

Together these effects make Cl's electron-gain enthalpy more negative than F's.

Video Solution:



Q7 Text Solution:

A.

$N < Be < B$

Correct: $Be \text{ (atomic)} > B \text{ (atomic)} > N \text{ (atomic)}$

B. $F^- < O^{2-} < N^{3-}$

All are isoelectronic (10 electrons).

Radius increases as nuclear charge decreases:

$F^- (Z=9) < O^{2-} (Z=8) < N^{3-} (Z=7)$; Correct.

C. $Na < Li < K$

Incorrect: $K > Na > Li$ (size increases down group)

D. $Fe^{3+} < Fe^{2+} < Fe^{4+}$

Fe^{4+} has higher charge $\rightarrow$ smaller size than Fe^{3+} .

Correct order: $Fe^{4+} < Fe^{3+} < Fe^{2+}$

Video Solution:



Q8 Text Solution:

- **N_2O_5 vs P_4O_{10} :** The oxide of nitrogen forms nitric acid (stronger) than phosphoric acid from phosphorus pentoxide, so N_2O_5 is actually more acidic.
- **CO_2 vs SiO_2 :** CO_2 is distinctly acidic in water (forms carbonic acid), whereas SiO_2 is largely insoluble and weakly acidic, so CO_2 is more acidic.
- **B_2O_3 vs BeO :** B_2O_3 is acidic; BeO is amphoteric, so B_2O_3 is more acidic.
- **$HClO_3$ vs $HClO_4$:** Perchloric acid ($HClO_4$) is stronger.

Video Solution:



Q9 Text Solution:

Ionization Potential (I.P.) = Energy required to remove an electron.

Series-I: $[N^{3-}, O^-, Na^+] \rightarrow$ Lowest I.P.

- **N^{3-} :**
 - Configuration: $[Ne]$ (10 electrons)
 - Charge: $-3 \rightarrow$ high electron repulsion $\rightarrow$ **easiest to remove electron**
 - **Lowest I.P.**



- O^- :
 - [Ne] but $Z=8 \rightarrow$ more nuclear pull than $N^{3-} \rightarrow$ higher I.P.
- Na^+ :
 - [Ne], $Z=11 \rightarrow$ cation $\rightarrow$ **highest I.P.** (hardest to remove electron)

$\rightarrow$ **Lowest I.P. in Series-I: N^{3-}**

Series-II: $[N^+, C^+, O^+] \rightarrow$ Highest I.P.

All are **isoelectronic** ([He] 2 electrons), but different nuclear charge:

Ion	Z	I.P. Trend
C^+	6	Lowest
N^+	7	Medium
O^+	8	Highest (highest nuclear charge)

$\rightarrow$ **Highest I.P. in Series-II: O^+**

Video Solution:



Q10 Text Solution:

Element	Neutral Atom	M^+ (after 1st I.P.)	Electron removed in I.P. ₂	
K	[Ar] $4s^1$	[Ar]	From noble gas core [Ar]	Very high $I. P_2$
Ca	[Ar] $4s^2$	[Ar] $4s^1$	From $4s^1$	Moderate $I. P_2$
Ba	[Xe] $6s^2$	[Xe] $6s^1$	From $6s^1$	Lowest $I. P_2$

Video Solution:



Q11 Text Solution:

Electronegativity difference ($\Delta\chi$) determines bond type:

- $\Delta\chi < 1.7 \rightarrow$ Covalent bond (polar or non-polar)
- $\Delta\chi > 1.7 \rightarrow$ Ionic bond
- $\Delta\chi = 1.7 \rightarrow$ Borderline (often considered start of ionic character)

Video Solution:



Q12 Text Solution:

Electron gain enthalpy can be exothermic or endothermic.

Order of ionic radius $O^{2-} > F^-$.

[New NCERT Class 11th Page No. 89]

Video Solution:



Q13 Text Solution:

Electron Affinity (EA): Energy released when an electron is added to a neutral atom in the gas phase. High EA means the atom strongly attracts an additional electron.

Ionization Energy (IE): Energy required to remove an electron from a neutral atom. High EA generally correlates with high IE because:



Both increase across a period (left to right) due to increasing nuclear charge and decreasing atomic size.

Atoms with high EA (e.g., halogens like Cl, F) also have high IE — they gain electrons easily but lose them with difficulty.

Video Solution:



Q14 Text Solution:

I. I.P of O(g) < I.P. of O⁻(g)

- O(g): [He] 2s² 2p⁴ → O⁺(g) + e⁻
- O⁻(g): [He] 2s² 2p⁵ → O(g) + e⁻
- Removing e⁻ from O⁻ (repulsion) is **easier** than from neutral O.
- → **I.P. of O(g) > I.P. of O⁻(g) → FALSE**

II. I.P of Ne(g) > I.P. of Ne⁺(g)

- Ne(g): [He] 2s² 2p⁶ → Ne⁺(g) + e⁻
- Ne⁺(g): [He] 2s² 2p⁵ → Ne²⁺(g) + e⁻
- Ne⁺ has **higher Z_{eff}** and **same shell** → **harder to remove e⁻**
- → **I.P. Ne(g) < I.P. Ne⁺(g) → FALSE**

III. E.A. of O⁺(g) > E.A. of O(g)

- O(g) + e⁻ → O⁻(g) → E.A. = -141 kJ/mol
- O⁺(g) + e⁻ → O(g) → **exothermic** (cation attracts e⁻ strongly) → **E.A. of O⁺ > E.A. of O** (by definition, more energy released) → **TRUE**

IV. I.P of N(g) > I.P. of N⁺(g)

- N(g): [He] 2s² 2p³ → N⁺(g) + e⁻
- N⁺(g): [He] 2s² 2p² → N²⁺(g) + e⁻
- N⁺ has **higher Z_{eff}**, removing from **2p²** (not half-filled) → **higher I.P.**
- → **I.P. N(g) < I.P. N⁺(g) → FALSE**

Video Solution:



Q15 Text Solution:

Electron Gain Enthalpy (EGE) Trends:

Noble gases (Ne, Ar):

Have positive electron gain enthalpy (they don't accept electrons easily).

Halogens (F, Cl):

Have very negative electron gain enthalpy (they strongly attract electrons).

2. Approximate EGE values (in kJ/mol):

Pair	Difference in EGE (Approx.)
Ne – F	116 – (-333) = 449
Ne – Cl	116 – (-349) = 465 (Maximum)
Ar – Cl	96 – (-349) = 445
Ar – F	96 – (-333) = 429

The difference is maximum between Ne and Cl.

Video Solution:



Q16 Text Solution:

- Lower Z_{eff} → Weaker attraction on valence electron
- → Easier to remove → IE decreases

Video Solution:



Q17 Text Solution:



Electron gain enthalpy generally becomes more negative across a period as we move from left to right. Within a group, electron gain enthalpy becomes less negative down a group. However, adding an electron to the 2p-orbital leads to greater repulsion than adding an electron to the larger 3p-orbital, hence the element with most negative electron gain enthalpy is chlorine: the one with the least negative electron gain enthalpy is phosphorus.

Video Solution:



Q18 Text Solution:

Bond	$\Delta E.N.$	Nature
M-O	Higher (if M is metal, $\chi_M < 1.5$)	More ionic
O-H	1.4	Polar covalent

→ If $\Delta(E.N.) \text{ M-O} > \Delta(E.N.) \text{ O-H} \rightarrow \text{M-O is more ionic} \rightarrow \text{breaks first}$

Video Solution:



Q19 Text Solution:

Be ($[\text{He}] 2s^2$) has a fully filled 2s subshell → higher stability and 2s electron is more penetrating (higher Z_{eff}). B ($[\text{He}] 2s^2 2p^1$) removes a 2p electron → higher energy, less penetrating → easier to remove. Hence, IE of B < Be due to 2p orbital of valence electron in B.

Video Solution:



Q20 Text Solution:

Big difference in 3rd and 4th ionisation energy. Therefore, valence electron in the element are three.

Video Solution:



Q21 Text Solution:

K_2O , $\text{SO}_3 \rightarrow \text{Basic}$
 P_2O_5 , Cl_2O_7 , $\text{SO}_3 \rightarrow \text{Acidic}$
 ZnO , SnO_2 , $\text{SO}_3 \rightarrow \text{Amphoteric}$

Video Solution:



Q22 Text Solution:

In general ionisation enthalpy increases along the period and decreases on moving down the group. So, the correct order for ionisation enthalpy is as follows : $\text{Zn} > \text{Fe} > \text{Cr} > \text{Sc}$.

Video Solution:



Q23 Text Solution:



In case of isoelectronic species:

$$\text{Ionic size} \propto \frac{-\text{ve charge}}{+\text{ve charge}}$$

Radii order of the species of an element:

$$A^- > A > A^+$$

Thus correct radii order are :

$$1) H^- > H > H^+$$

$$2) Na^+ < F^- < O^{2-}$$

$$3) O^{2-} > F^- > Na^+$$

$$4) Al^{3+} < Mg^{2+} < N^{3-}$$

Video Solution:



Q24 Text Solution:

$$\sigma \text{ for Na} = 0 \times 0.35 + 8 \times 0.85 + 2 \times 1$$

$$= 0 + 6.8 + 2 = 8.8$$

$$Z_{\text{eff}} = Z - \sigma$$

$$Z_{\text{eff}} = 11 - 8.8 = 2.2$$

$$\sigma \text{ for Na}^+ = 7 \times 0.35 + 2 \times 0.85$$

$$= 2.45 + 1.7 = 4.15$$

$$Z_{\text{eff}} = Z - \sigma$$

$$= 11 - 4.15 = 6.85$$

Video Solution:



Q25 Text Solution:

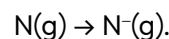
Sum of $\Delta_f H_1$ and $\Delta_f H_2$ is minimum for element III so, it forms divalent cation easily and form stable binary halide AX_2 .

Video Solution:



Q26 Text Solution:

Adding an electron to a nitrogen atom is endothermic (positive electron-gain enthalpy), so energy must be absorbed for



For F, Cl and H the first electron addition is exothermic.

Video Solution:



Q27 Text Solution:

Electronegativity increases as we move left to right in a period and decreases as we move down in the group.

(i) C and N belong to the same period. Hence N has more electronegativity than carbon (C).

(ii) In the same way Si is more electronegative than C.

(iii) As Si is below the carbon in 14th group the electronegativity of carbon is more than of Si.

Video Solution:



Q28 Text Solution:

The first three ionisation energies rise modestly (100 eV $\rightarrow$ 150 eV $\rightarrow$ 181 eV), but a huge jump occurs at the 4th (2000 eV).

This shows the element readily loses **three** outer-



shell electrons, after which an inner closed shell is disturbed. Such a pattern is characteristic of **Group 13 (III-A)** elements.

- **Group 13 elements are metallic** (except B which is metalloid) → statement A.
- Their common oxide formula is A_2O_3 → statement B.
- Oxides like Al_2O_3 (and those of Ga, In) are **amphoteric** → statement C.

Hence all given statements are correct.

Video Solution:



Q29 Text Solution:

The atomic number 119 corresponds to the first element in the hypothetical 8th period of the periodic table. Elements that are the first in their period, such as hydrogen, lithium, and sodium, belong to Group 1, the alkali metals. Therefore, the group number x is 1.

Elements in Group 1 have a single valence electron. They tend to lose this electron to form a stable positive ion with a charge of +1. The valency is the number of electrons an atom can lose, gain, or share. In this case, the valency y is 1.

Video Solution:



Q30 Text Solution:

$x = 5$ ($CrO_3, Mn_2O_7, N_2O_3, SO_2, CO_2$)

$y = 3$ (Na_2O, CaO, V_2O_3)

$z = 4$ (amphoteric), all other

$x^2 + 2y + z = 25 + 6 + 4 = 35$

Video Solution:



Q31 Text Solution:

In a 9-period periodic table, different subshells (s, p, d, f, g, h) are considered.

Step-by-step:

- Each subshell has a certain number of orbitals:
 $s = 1, p = 3, d = 5, f = 7, g = 9, h = 11 \rightarrow \text{Total} = 36$ orbitals
- Each orbital can accommodate 5 electrons (assuming 5 periods overlapping).
- Total electrons = 36 orbitals $\times$ 5 electrons = 180 electrons
- Since each electron corresponds to one element (unique configuration),
 $\rightarrow 180 \text{ electrons} = 180 \text{ elements}$

Video Solution:



Q32 Text Solution:

$X \Rightarrow \frac{1s^2}{K} \Rightarrow \text{Value of } n \text{ is '6'}$

Configuration of 'X' = $1s^2 \dots \dots \dots$

$4s^2 4p^6 4d^{10} 4f^7 5s^2 5p^6 5d^0 6s^2$

No. of unpaired electrons = 7

Video Solution:



Q33 Text Solution:



Representative elements are those which contain only s and p-block elements.

Video Solution:



Q34 Text Solution:

$Z = 104$ (Rf)
 $[Rn]5f^{14}6d^27s^2$
 Period no. = $7 = x$
 Group no. = $4 = y$
 $x + y = 11$

Video Solution:



Q35 Text Solution:

$$\begin{aligned}\Delta EN &= |2.5 - 2.1| \\ &= 0.4 \\ \%IC &= 16|\Delta EN| + 3.5|\Delta EN|^2 \\ &= 16 \times 0.4 + 3.5 \times 0.4^2 \\ &= 6.96\% \\ &\approx 7\%\end{aligned}$$

Video Solution:



Q36 Text Solution:

Aluminium and gallium oxides are amphoteric in nature.

$X = Al_2O_3, Ga_2O_3, Al(OH)_3, Ga(OH)_3$

$Y = In_2O_3, Tl_2O_3, In(OH)_3, Tl(OH)_3$

Video Solution:



Q37 Text Solution:

Ionization energy of Cl is high due to its high nuclear charge and small size. Compare each:

- $Na < Cl$
- $Mg < Cl$
- $P < Cl$
- $S < Cl$
- $Ar > Cl$
- $Al < Cl$

So, total elements with ionization energy less than Cl = 5 (Na, Mg, P, S, Al).

Final Answer: 5.

Video Solution:



Q38 Text Solution:

Electronic configuration that follow $(n + l)$ rule is as

$$(1s^2) (2s^2 2p^6) (3s^2 3p^6) (4s^1)$$

$$\text{So, } \sigma = 8 \times 0.85 + 10 \times 1.00 = 16.80$$

$$z_{eff} = 19 - 16.80 = 2.20$$

$$x = 2.20$$

Electronic configuration does not follow $(n + l)$ rule is as

$$(1s^2) (2s^2 2p^6) (3s^2 3p^6) 3d^1$$

$$\text{So, } \sigma = 18 \times 1.00 = 18.00$$

$$z_{eff} = 19 - 18 = 1$$

$$y = 1$$

$$x + y = 2.20 + 1 = 3.20 \approx 3$$

Video Solution:



**Q39 Text Solution:**

Given:

Mass = 10 g, Molar mass = 30 g/mol

Moles = $10 / 30 = 1/3$ mol

Electron gain enthalpy = $-3 \text{ eV/atom} = -3 \times 23 = -69 \text{ kcal/mol}$

Total energy released = $(1/3) \times (-69) = -23 \text{ kcal}$

Energy released = 23 kcal/mol

Final answer: 23

Video Solution:**Q40 Text Solution:**

The minimum amount of energy which is required to remove the most loosely bound electron from neutral gaseous atom is known as ionisation energy.

$N_2 > N$ (1st I.E., because in N_2 last e^- remove from bonding M.O.)

$O_2 < O$ (1st I.E., because in O_2 last e^- remove from antibonding M.O.)

$H_2 > H$ (1st I.E., last e^- from B. M.O.)

$N_2^+ > N_2^-$ (stability, correct)

$O_2 > O$ and $H_2 > H$ (1st electron affinity, incorrect)

The thermal stability of group 1 (alkali metal) carbonates generally increases as you move down the group.

P_4O_{10} has 4 co-ordinate bond.

P_4O_8 has 2 π -bond.

Video Solution:



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